

1 Introduction

Your photos are sitting in a foreign data centre right now. Your device runs out of space, and the only two buttons on screen are *delete* or *pay*. That message is not an accident: it is a business model, and it has quietly become the default way the world stores its memories.

SANDOOQ is a Saudi-based, self-hosted cloud, very similar to built-in iCloud — except the server is yours. Files are encrypted end-to-end with Cryptomator (AES-256), nodes stay private over Tailscale and SSH, so even the operator cannot peek. Privacy you can prove, seamless backup, private file sharing on any device, extra storage for free. Decentralized by design, with a potential business model that lets anyone rent out spare space and earn. Your cloud. Your rules.

70%

of iPhone users who run out of space pay for more

~2/3

of US iCloud users pay for extra storage — more than any other Apple service

\$70/yr

average cost of a paid cloud plan on your phone

85%

of world data-centre power sits in the US, Europe and China

The three stacked problems

- **The capacity trap.** Your device fills up, and the paid upgrade is framed as the only answer.
- **The ownership gap.** You pay for the storage but never own it, and you can never switch it off.
- **The sovereignty gap.** Your files are copied to foreign data centres under laws you never agreed to.

Why it matters here. The Kingdom's cloud market is projected to grow from USD 5.07 bn (2025) to USD 14.61 bn (2030), while CST and NCA rules require sensitive data to stay inside the Kingdom. Yet most personal data still lives abroad.

2 Methodology

The proof of concept is a working two-node system: a rented server with a public IP, and a Mac at home with no public address. Both join one private Tailscale tailnet (WireGuard), so administration, federation and backups travel on an encrypted private network while the public internet sees only HTTPS on port 443.

SANDOOQ connection framework

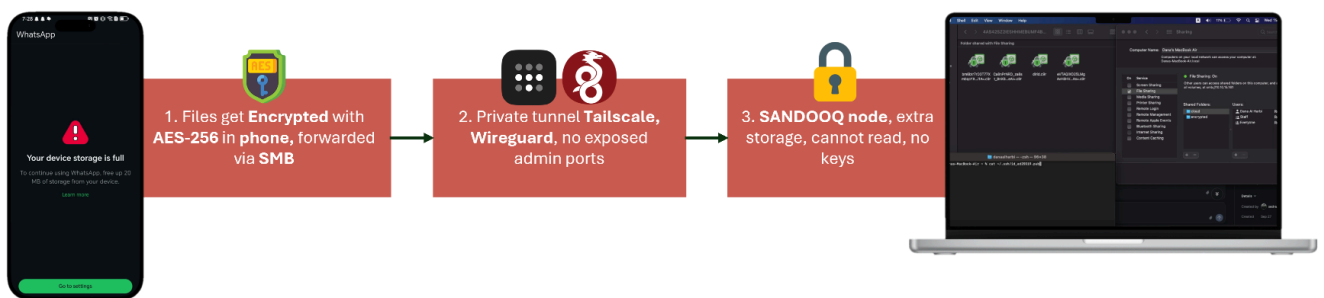


Figure 1 — SANDOOQ connection framework. A file is encrypted with AES-256 on the phone, forwarded over a private Tailscale/WireGuard tunnel with no exposed admin ports, and stored on a node that holds no keys and cannot read it.

SANDOOQ storage types



2.1 The technology, concretely

Component	Role in the project
Cryptomator (AES-256)	Encrypts files on the device before upload; the node stores unreadable .c9r blocks.
Tailscale / WireGuard	Private, device-authenticated tailnet joining both nodes and the operator's phone (stable 100.x.y.z).
SSH (Ed25519) + Termius	Key-only administration on a custom port; the private key never leaves the device.
Nextcloud / Seafile	Cloud layer on the public node: accounts, quotas, sharing, federation.
Samba (SMB) + Apple Files	Native iOS access to storage nodes; SFTP via Secure ShellFish / Owlfiles.
Let's Encrypt + Docker/snap	Trusted HTTPS on the public node and repeatable, isolated deployment.

2.2 Layers of protection — four independent walls

Layer	What it protects	How
Network access	Who can even reach a port	Tailscale tailnet, WireGuard, device authentication and access rules
In transit	Data while it moves	HTTPS/TLS for users; SSH or WireGuard for operators
At rest	Data sitting on a disk	Full-disk encryption plus the encrypted vault itself
End to end	Data from everyone, including the operator	Cryptomator vaults; the key is created on the user's device and never transmitted

3 Results

The system was built and tested end-to-end from the phone and the server. The headline outcomes are measurable, not descriptive:

2

nodes joined on one encrypted tailnet (public server + home Mac)

0

readable bytes for the operator — disk shows only encrypted .c9r names

Instant

encrypted vault round-trip verified from phone and server

0 SAR

paid to a foreign provider for the demo storage

- Working prototype across a public node and a home node via SMB and the tailnet.
- On-disk files verified unreadable; only encrypted .c9r names are visible.
- AES-256 encrypted vault round-trip confirmed from the phone and the server.

4 Innovation

FAMILIAR

Feels like iCloud

Open Files on any device and the files are simply there.

PRIVATE

Server cannot read

Encrypted on your device before it travels anywhere.

REACHABLE

Private tunnel

Phone, Mac and server on one encrypted tailnet; no public admin ports.

SOVEREIGN

Yours and in the Kingdom

Hosted and administered in Saudi Arabia; anyone can run a node and earn.

5 Security architecture: two nodes, one private network

Node	Exposure
Public node	Rented server, public IP. HTTPS 443 for users; SSH closed to the internet.
Private node	Mac at home, no public address. Reachable only inside the tailnet (100.x.y.z).

Two layers, both required. Cryptomator protects file *contents* from everyone, including the operator. Tailscale protects the *network path* and removes unnecessary exposure. Neither replaces the other.

What Tailscale changes

- **Smaller attack surface.** The SSH port can be closed to the whole internet, so scanners have nothing to reach.
- **The private node becomes useful.** The home Mac is reachable from anywhere with no ports opened on the router.
- **Federation becomes real.** Two nodes on one private network can federate and back each other up.
- **Identity, not just secrecy.** Every device authenticates before joining and can be revoked instantly.

Who can read what

Actor	Can read?
The user	Yes — holds the key
Node operator / host	No
Network eavesdropper	No
Internet port scanner	Nothing to reach
Other users	No
Thief with the disk	No

Stated plainly: a host can observe metadata, and whoever controls a disk can delete data. What no one can do is *read* it — and that claim is testable.

6 A federated mesh across the Kingdom

SANDOOQ is not one central server with smaller servers attached. It is a mesh of equal peer nodes, each independently operated, joined through federation. Nodes reach each other inside a private tailnet rather than over the open internet: there is no centre, and nothing for a stranger to scan.

SUPPLY

Students, families and small businesses with idle storage install the host app, declare capacity, pass a readiness check, and earn SAR monthly for the space actually used.

DEMAND

Phone users, families and schools need space. They pick a node by region, price, latency and trust — and are never locked to one operator.

SANDOOQ

Provides encryption, accounts, trust, matching and payouts, and takes a small commission on each rental.

Economics (illustrative)

Host offers 500 GB on an idle drive	40% rented = 200 GB
Market price SAR 10 / 100 GB / month	Gross SAR 20 / month
Platform commission 20%	SAR 4
Host payout	SAR 16 / month from hardware earning nothing

Illustrative figures: they show the shape of the model, not a promise of income. Today 100% of a foreign subscription leaves the Kingdom; under SANDOOQ the same payment is redistributed locally.

7 Challenges faced, and how they were solved

Challenge	How it was solved
Tooling friction: no visible key manager, non-default SSH port.	Generated an Ed25519 key in macOS Terminal with ssh-keygen and used port 52156 explicitly. Both became deliberate security choices.
iPhone Files app refused to connect.	Discovered it speaks SMB only (AFP is retired). Gave Files an SMB / file-provider route and kept SSH for administration.
Private node only worked on home Wi-Fi.	Joined both nodes and the phone to a Tailscale tailnet; the home Mac gained a stable 100.x.y.z address and the server's SSH port could then be closed.
Cryptomator "looked broken" twice.	Learned a vault protects data from storage and others, not from the key holder, and only while files are inside it. Verified by seeing scrambled names without unlocking.
No way to prove privacy to judges.	Built a repeatable demo: encrypt on device, show the storage folder, display only unreadable blocks.

8 Why SANDOOQ is better than iCloud

Dimension	iCloud	SANDOOQ
Who can read your files	Apple holds the keys; you trust a policy.	Only you; the key never leaves your device.
Where your files live	Data centres abroad.	Inside Saudi Arabia.
Can you verify privacy	No — you take their word.	Yes — the disk shows unreadable encrypted blocks.
What you pay	A subscription, forever.	A free tier, then plans priced below mainstream providers.
Who earns from your storage	A foreign corporation.	Saudi hosts, families and small businesses.
Economic effect	Money, data and skills leave the country.	Money, data and skills stay in the Kingdom.

In one line. iCloud rents you space on someone else's computer. SANDOOQ gives you encrypted space on infrastructure you and your neighbours own — and pays those neighbours for it.

9 Privacy, cryptographically

STEP 1

Your device

The file is encrypted with your key before it travels.

STEP 2

Encrypted tunnel

HTTPS for users; the tailnet for operators.

STEP 3

Storage node

Stores unreadable blocks — no key, no way to read.

STEP 4

Only your device

Can decrypt the file again.

This is the actual content of the vault on the storage:

```
dirid.c9r
bm6btr7rD3T77XmbqcYX...i1A=.c9r
eVTAQlX025LMgAvH8HV...As=.c9r
Ca8nPRNRD_za8st_8nXB...eAl=.c9r
```

No readable names, no images, no text, no clue what any file contains. This is what the storage — and anyone who reaches it — actually sees. Only the user's device can turn it back into files.

10 Market, rollout and risks

First users

Students with full phones and no budget, families with scattered photos, privacy-focused users, small businesses needing local storage, and hosts who want idle hardware to pay for itself.

Channels

Short-form video showing a photo turning into unreadable data; campus ambassadors; gaming and creator communities; national technology events such as LEAP and SDAIA initiatives; device retailers and ISPs as host onboarding points; referral programme; plain Arabic and English education content.

Rollout phases

0 • Proof of concept (now)	Two nodes, HTTPS users, accounts, quotas, vault, host demo.
1 • Private pilot (1-3 mo)	10–30 testers in the tailnet, onboarding, backups, monitoring.
2 • Multi-user product (3-6 mo)	Self-service signup, 2FA, mobile apps, billing, support.
3 • Federation (6-12 mo)	Federated nodes, node directory, in-app node picker.
4 • Market launch (12-18 mo)	Host app, reputation, payouts, commission.
5 • Scale (18 mo+)	National network, B2B and white-label, partnerships, compliance.

Risks and responses

Risk	Response
Exposed administration	SSH closed to the internet, reached only inside the tailnet.
Security attacks	HTTPS everywhere, key-only SSH, updates, firewall, 2FA, least privilege, per-device access rules.
Abuse and data loss	Clear terms, account limits, takedown process, backups in a second location with tested restores.
Untrustworthy hosts	End-to-end encryption removes snooping; verification tiers and public reputation manage reliability.
Regulation	In-Kingdom hosting, documented policy, minimal metadata, clear consent under PDPL.

11 Future work

NEXT

Federated node + directory

A second node and an in-app node directory.

NEXT

Minimum host earning flow

Register capacity, accept encrypted writes, see usage.

NEXT

One-command installer

So anyone can reproduce the setup.

NEXT

Post-quantum encryption

Future-bulletproof the vault against quantum attacks.

SANDOOQ. Storage you own. Privacy you can verify. Income that stays in the Kingdom.

Saudi-based, self-hosted and encrypted end-to-end, administered over a private Tailscale network, and built so that the people who provide the storage are the people who earn from it.

References

1. All About Cookies, *The Cost of Running Out of Space* (survey), 2026.
2. Consumer Intelligence Research Partners (CIRP), US iCloud subscription survey, 2024.
3. MarketsandMarkets, *Cloud Storage Market — Global Forecast to 2031*, 2026.
4. International Energy Agency (IEA), *Energy and AI*, 2025.
5. MarketsandMarkets, *Saudi Arabia Cloud Computing Market — Forecast to 2030*, 2025.
6. CST and NCA regulations on data residency in the Kingdom of Saudi Arabia.